Yuanhao Chen

Ph.D. Student in Mechanical Engineering, New Jersey Institute of Technology

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Research Interests

Rehabilitation robotics, wearable exoskeletons, human biomechanics and dynamic stability, multimodal sensing, human-robot interaction

Education

New Jersey Institute of Technology — Newark, NJ

Ph.D. in Mechanical Engineering | AnyMotionLab — *Aug. 2025 – Present*

Arizona State University — Tempe, AZ

M.S. in Mechanical and Aerospace Engineering | GPA: 3.83/4.00 — *Jan. 2023 – May 2025*

- Thesis: *Evaluating the Impact of a Back Support Device on Human Stability During Trip Perturbations*

Southern Illinois University Edwardsville — Edwardsville, IL

B.S. in Mechatronics and Robotics Engineering | GPA: 3.42/4.00 — *Aug. 2019 – May 2022*

- University Dean's List; Member, Honor Society of Phi Kappa Phi

Research Experience

New Jersey Institute of Technology | *AnyMotionLab* — Aug. 2025 – Present

- Developing a home-based upper-limb rehabilitation system for stroke survivors that combines a wearable exoskeleton, multimodal sensing and serious games.
- Leading field usability studies of the system with older adults at community senior centers.
- Led a study of walking stability during load carrying with commercial shoulder-assist exoskeletons.
- Leading a usability study of a commercial robotic hand glove and a survey study on workers' perceptions of occupational exoskeletons.

Arizona State University | *Neuromuscular Control and Human Robotics Laboratory* — May 2023 – May 2025

- Designed an experimental framework to evaluate how a soft, variable-stiffness back-support device affects postural stability during trip-like perturbations and daily tasks.
- Quantified the device's biomechanical impact with motion capture and statistical analysis, relating device stiffness to dynamic balance.

Xi'an Jiaotong University | *Wearable Health Monitoring Team* — 2022 – 2023

- Developed a Shallow ConvNet model to classify epileptic electroencephalogram (EEG) signals.
- Developed a control method to improve the performance of nonlinear motor systems.

Publications

Journal Articles

[J1] R. Khatavkar, T. B. Nguyen, **Y. Chen**, H. Lee, J. Sun, "A Hybrid Variable-Stiffness Soft Back Support Device," *IEEE Robotics and Automation Letters*, vol. 10, no. 4, Apr. 2025.

Conference Proceedings

[C1] **Y. Chen**, N. Ratnakumar, M. Rodriguez, M. Tohfafarosh, L. Zheng, C. Pan, X. Zhou, W. Yin, "Exploring Effects of Shoulder-Assist Exoskeletons on Load-Carrying Stability: A Phase-Specific Approach," *Proc. 70th HFES International Annual Meeting (ASPIRE 2026)*, Reno, NV, Oct. 2026. doi: 10.1177/10711813261475219

[C2] **Y. Chen**, R. Khatavkar, S. Nayak, J. Sun, H. Lee, "Impact of a Soft Wearable Back-Support Device on Postural Stability during Trip-Like Perturbations," *IEEE International Conference on Biomedical Robotics and Biomechatronics (BioRob 2026)*, Aug. 2026.

[C3] **Y. Chen**, M. Rodriguez, M. Tohfafarosh, L. Zheng, X. Zhou, W. Yin, "Investigating Walking Stability of Exoskeleton-Assisted Load Carrying Tasks: A Pilot Comparison of Three Commercial Devices," *IISE Annual Conference & Expo 2026*, May 2026.

[C4] **Y. Chen**, Y. Xu, X. Wang, et al., "Nonlinear Vibration and Bifurcations of a Bending-Twisting Rotor System," *ASME International Mechanical Engineering Congress and Exposition (IMECE 2024)*, Nov. 2024.

[C5] **Y. Chen**, Y. Xu, X. Wang, "A Novel Model for Heart Beating Rhythms," *ASME International Mechanical Engineering Congress and Exposition (IMECE 2024)*, Nov. 2024.

Other Publications

[O1] **Y. Chen**, Z. Zhan, "Clairvoyance: Vision-Impaired Friendly Assistive Mobile Device," *Applied and Computational Engineering*, Feb. 2024.

Conference Presentations

- Oral presentation, *ASPIRE 2026, 70th HFES International Annual Meeting*, Session "Exoskeletons in Occupational Applications," Reno, NV, Oct. 2026.
- Presentation, *IISE Annual Conference & Expo 2026*, May 2026.
- Presentation, *ASME International Mechanical Engineering Congress and Exposition (IMECE 2024)*, Nov. 2024.

Innovation and Entrepreneurship

- **Entrepreneurial Lead**, NSF I-Corps, Northeast Region Hub, Fall 2026: customer discovery for home-based upper-limb rehabilitation.

Professional Service

- **Reviewer**, IISE Annual Conference & Expo 2026
- **Reviewer**, ASME International Mechanical Engineering Congress and Exposition (IMECE) 2024

Honors and Awards

- Member, Honor Society of Phi Kappa Phi 2022
- University Dean's List, Southern Illinois University Edwardsville 2019 – 2022

Selected Projects

- **Learned sampling for RRT* path planning** (2023): conditional VAE trained on generated mazes to bias RRT* sampling, giving lower path cost and higher success rate than uniform sampling.
- **Clairvoyance assistive device** (2022): GPS module with sub-10 cm accuracy and camera-based recognition of traffic lights and crosswalks for visually impaired users.
- **CIM conveyor operating system** (2022): conveyor line, PLC ladder logic and a Visual Basic / HMI operator interface built from scratch.

Technical Skills

Biomechanics and sensing: OptiTrack Motive, Vicon Nexus, OpenSim, wearable IMUs, eye-tracking and depth-camera data acquisition, Lab Streaming Layer

Programming: Python, MATLAB, C/C++, C#, R

Development: Unity, ANSYS, Arduino, LabVIEW, PLC, PSoC Creator, LaTeX

Languages: English (professional), Mandarin Chinese (native)